

Signal Distortion, Feedback Loops, and Reality Drift

How Optimized Systems Amplify the Wrong Signals

A. Jacobs | Reality Drift Framework

The Absence of Failure examines systems that remain active, orderly, and apparently successful even as their corrective contact with reality weakens. Each paper traces a different mechanism through which internal evidence of success can continue accumulating after alignment with the underlying conditions has begun to erode.

The Basic Pattern

Modern information systems rely on signals to determine what content spreads and what disappears. Platforms must decide which posts to promote, which videos to recommend, and which stories to surface to large audiences. Because meaning, quality, and truth are difficult to measure directly at scale, platforms instead rely on behavioral signals. Examples include:

- clicks as a proxy for interest
- shares as a proxy for importance
- watch time as a proxy for value
- engagement as a proxy for relevance

At first, these signals function as useful indicators. They help platforms identify content that appears to resonate with audiences. As optimization intensifies, however, a familiar pattern can emerge. Algorithms increasingly prioritize the signals themselves rather than the underlying informational value those signals were meant to represent.

Signal distortion occurs when systems optimize behavioral signals faster than those signals remain aligned with real informational value.

When the System Produces Its Own Evidence

The platform remains coherent because ranking, visibility, creator behavior, and audience response continually reinforce the signals the system was designed to reward. Amplification generates attention, that attention produces engagement, and the resulting engagement is then treated as evidence that further amplification was justified.

The system therefore begins producing part of the evidence it uses to determine what deserves visibility.

When Signals Become Targets

Signal-based systems begin to drift when the indicators used to measure attention become the primary targets of optimization. Engagement begins to stand in for informational value, virality for importance, emotional intensity for relevance, and visibility for substance.

As this shift occurs, the signal stops functioning merely as a measure of audience response and becomes a target of optimization itself. Content increasingly evolves to maximize the signals that drive algorithmic distribution, producing a feedback loop in which the most rewarded signals become the most visible and the most visible signals become the most rewarded.

Over time, this process can reshape the information environment itself.

Related Concepts Across Fields

Different fields describe this pattern using different terminology. Media researchers often refer to algorithmic amplification, where recommendation and ranking systems disproportionately promote content based on engagement signals. Economists and platform analysts describe related dynamics through the attention economy, where competition for user attention drives optimization toward increasingly stimulating material. Technology critics often describe the outcome as engagement optimization, where systems prioritize measurable interactions such as clicks, shares, watch time, and reactions.

Although these concepts emphasize different aspects of the process, they describe the same structural pattern. Optimization shifts toward signals that are easier to measure than the informational value those signals were originally intended to represent.

How Signal Distortion Emerges

The shift from signal measurement to signal distortion typically unfolds in several stages.

Stage 1 - Signal Measurement

Platforms introduce metrics that track how users interact with content. These signals provide a simplified way to estimate interest, relevance, or value and help guide early recommendation systems.

Stage 2 - Algorithmic Optimization

Recommendation algorithms increasingly optimize for signals that correlate with engagement. Content that generates stronger measurable reactions becomes more likely to receive additional distribution.

Stage 3 - Feedback Loop Amplification

As algorithms repeatedly promote content that produces the strongest signals, creators adapt their behavior to trigger those same signals. This may include:

- emotionally intense headlines
- outrage-driven commentary
- simplified narratives
- content designed for maximum shareability

The environment begins adapting to the logic of the ranking system, and content that produces stronger reactions becomes increasingly likely to spread.

Stage 4 - Signal Dominance

Eventually, the signals themselves begin shaping the broader information environment. Content that generates measurable reactions receives greater visibility than material optimized for accuracy, depth, or nuance, allowing the logic of engagement to influence what kinds of information are produced and encountered.

Examples Across Platforms

Signal distortion appears across many digital information environments.

Social Media Feeds

Platforms such as TikTok, Instagram, and Facebook use behavioral signals such as watch time, likes, shares, and other interactions to help rank content. Material that generates strong measurable reactions can therefore receive greater distribution, which in turn creates additional opportunities for engagement.

Video Recommendation Systems

Video platforms use signals such as watch time and viewer retention to inform recommendation systems. Creators may adapt by producing increasingly attention-grabbing formats designed to maintain engagement and satisfy the signals that influence distribution.

News and Media Ecosystems

Online media environments frequently reward stories that generate strong reactions. Outrage, controversy, and emotionally intense narratives can attract more attention than slower or more nuanced reporting, creating incentives for publishers to adapt content to the signals that drive circulation.

Creator Economies

Influencers and other content creators learn which formats, topics, and presentation styles receive distribution. Over time, this can encourage content designed primarily to maximize visibility and engagement rather than informational value.

Signal Distortion and Reality Drift

Within the Reality Drift framework, signal distortion shows how an information system can remain highly active and apparently responsive while becoming progressively less reliable as a representation of the world it is filtering.

Engagement rises, virality accelerates, and content circulates continuously, yet those indicators increasingly reflect the behavior of a system organized around producing and amplifying them. Internal evidence of success can therefore grow even as the relationship between visibility and informational value weakens. What appears to be evidence of relevance may increasingly be evidence of successful amplification.

Recursive Confirmation

Algorithms do not merely measure an independent information environment. Ranking changes what becomes visible, creators adapt to what receives distribution, and audiences encounter material that has already survived those filters. The resulting engagement is then fed back into the system and treated as evidence of relevance.

The system is therefore measuring an environment it has helped produce. As the cycle repeats, amplification can appear to validate the signals that originally caused the amplification.

Recognizing the Pattern

Signal distortion can remain difficult to detect because the system continues performing well according to its own internal indicators. Engagement remains healthy, activity stays high, and content circulates continuously even as the most visible material increasingly favors emotional intensity, simplification, or other characteristics rewarded by the distribution system.

The system can therefore appear highly active and responsive while becoming progressively less reliable at distinguishing what is valuable from what is merely effective at generating measurable reactions. Understanding signal distortion helps explain how optimized information systems can become increasingly successful at producing attention while drifting away from the informational qualities those signals were originally intended to approximate.

The question is not whether the system remains coherent. The question is what can still correct it.

The absence of failure is not proof of alignment. Coherence is not contact.